Supplementary Materials for Latent Variable Moderated Mediation Using Correct and Misspecified Models

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Realized Reliability and Explained Variance

The indicator error variances were calibrated under normality to the target reliability and then held fixed across the predictor distributions, and the structural disturbance variances were set so that $R^2 \approx 0.3$ for M and Y under normality, correct specification, and $a_3 = 0.2$. Because the variance of the product term and of M depends on the predictor distribution and on a_3, the realized reliability of the M and Y indicators and the explained variance of both equations vary across conditions. Table 1 and Table 2 report their realized values, computed in every generated dataset as the mean over the four indicators of $\lambda_j^2 \operatorname{Var}(\eta)/\operatorname{Var}(z_{\eta j})$ and as the variance of the structural part over the variance of the latent outcome, and averaged over replications and sample sizes. Under the cross-loading, the m_3 indicator carries variance from X that the formula does not count as true-score variance, so the reliability of M falls by construction. Under c_2 and c_3, the Y indicator error variances were rescaled to keep the reliability of Y at its baseline value, so only its R^2 rises. The residual covariance leaves both quantities unchanged. Each tabled value is a mean over 4,000 replications in Table 1 and twice that in Table 2, with its Monte Carlo standard error in parentheses, at most 0.0012 for the reliabilities and 0.0018 for the R^2 values.

Table 1

Realized Reliability of the Indicators by Distribution, Interaction Magnitude, and Reliability Level

Reliability	Distribution	a_3	X	W	M	Y	M (cl.s)	M (cl.m)
Low	Normal	0.0	.503 (.0005)	.503 (.0005)	.503 (.0005)	.503 (.0005)	.477 (.0005)	.442 (.0005)
		0.2	.503 (.0005)	.503 (.0005)	.502 (.0006)	.504 (.0005)	.477 (.0005)	.443 (.0005)
		0.4	.503 (.0005)	.502 (.0005)	.501 (.0006)	.502 (.0005)	.478 (.0005)	.444 (.0005)
	Uniform	0.0	.504 (.0004)	.504 (.0004)	.503 (.0005)	.503 (.0005)	.476 (.0005)	.441 (.0005)
		0.2	.504 (.0004)	.503 (.0004)	.497 (.0005)	.501 (.0005)	.471 (.0005)	.438 (.0005)
		0.4	.503 (.0004)	.503 (.0004)	.489 (.0005)	.498 (.0005)	.466 (.0005)	.433 (.0005)
	t_5	0.0	.501 (.0008)	.500 (.0008)	.502 (.0006)	.503 (.0005)	.477 (.0005)	.442 (.0005)
		0.2	.499 (.0008)	.500 (.0008)	.491 (.0006)	.497 (.0006)	.467 (.0006)	.435 (.0006)
		0.4	.501 (.0008)	.499 (.0007)	.490 (.0008)	.497 (.0006)	.472 (.0009)	.440 (.0008)
	χ^2 (s)	0.0	.500 (.0009)	.500 (.0008)	.501 (.0005)	.502 (.0005)	.477 (.0005)	.442 (.0005)
		0.2	.499 (.0008)	.500 (.0009)	.516 (.0007)	.507 (.0006)	.488 (.0006)	.453 (.0005)
		0.4	.499 (.0008)	.500 (.0008)	.525 (.0008)	.513 (.0006)	.496 (.0008)	.460 (.0007)
	χ^2 (d)	0.0	.498 (.0009)	.498 (.0008)	.502 (.0006)	.502 (.0005)	.477 (.0005)	.443 (.0005)
		0.2	.500 (.0009)	.499 (.0008)	.510 (.0006)	.507 (.0005)	.481 (.0005)	.447 (.0005)
		0.4	.499 (.0009)	.499 (.0009)	.514 (.0006)	.510 (.0006)	.483 (.0006)	.447 (.0005)
High	Normal	0.0	.702 (.0005)	.702 (.0005)	.701 (.0005)	.702 (.0005)	.663 (.0005)	.612 (.0005)
		0.2	.702 (.0005)	.702 (.0005)	.702 (.0005)	.702 (.0005)	.662 (.0005)	.613 (.0005)
		0.4	.703 (.0005)	.703 (.0005)	.702 (.0006)	.702 (.0005)	.663 (.0005)	.615 (.0005)
	Uniform	0.0	.703 (.0005)	.703 (.0005)	.702 (.0005)	.702 (.0005)	.662 (.0005)	.612 (.0005)
		0.2	.703 (.0005)	.704 (.0005)	.695 (.0005)	.700 (.0005)	.658 (.0005)	.608 (.0005)
		0.4	.703 (.0005)	.703 (.0005)	.690 (.0005)	.698 (.0005)	.654 (.0005)	.607 (.0005)
	t_5	0.0	.700 (.0007)	.697 (.0007)	.702 (.0005)	.702 (.0005)	.661 (.0005)	.612 (.0005)
		0.2	.699 (.0007)	.699 (.0007)	.691 (.0006)	.698 (.0005)	.656 (.0006)	.608 (.0006)
		0.4	.699 (.0007)	.699 (.0007)	.692 (.0007)	.697 (.0006)	.660 (.0007)	.617 (.0008)
	χ^2 (s)	0.0	.699 (.0008)	.698 (.0008)	.702 (.0005)	.703 (.0005)	.662 (.0005)	.613 (.0005)
		0.2	.698 (.0008)	.699 (.0008)	.713 (.0006)	.705 (.0005)	.671 (.0005)	.621 (.0005)
		0.4	.698 (.0008)	.698 (.0008)	.719 (.0007)	.709 (.0006)	.676 (.0007)	.625 (.0006)
	χ^2 (d)	0.0	.699 (.0008)	.698 (.0008)	.703 (.0005)	.702 (.0005)	.663 (.0005)	.612 (.0005)
		0.2	.698 (.0008)	.698 (.0008)	.710 (.0006)	.705 (.0005)	.665 (.0005)	.614 (.0005)
		0.4	.698 (.0008)	.697 (.0008)	.711 (.0006)	.708 (.0005)	.665 (.0005)	.612 (.0005)

Note. Mean reliability of the four indicators of each latent variable, averaged over replications and sample sizes, under the correctly specified model, with the Monte Carlo standard error in parentheses. The last two columns give the reliability of M under the small and medium cross-loading, the only (mis)specification that changes it. Targets are .5 (low) and .7 (high).

Table 2

Explained Variance of the Mediator and Outcome Equations by Distribution, Interaction Magnitude, and Structural (Mis)specification

Distribution	a_3	R^2_M	R^2_Y none	R^2_Y c_2.s	R^2_Y c_2.m	R^2_Y c_3.s	R^2_Y c_3.m
Normal	0.0	.276 (.0004)	.296 (.0004)	.413 (.0005)	.593 (.0006)	.340 (.0005)	.471 (.0007)
	0.2	.302 (.0005)	.301 (.0004)	.417 (.0005)	.595 (.0006)	.375 (.0005)	.527 (.0007)
	0.4	.368 (.0006)	.319 (.0005)	.430 (.0006)	.603 (.0006)	.416 (.0006)	.581 (.0008)
Uniform	0.0	.276 (.0004)	.296 (.0004)	.413 (.0005)	.593 (.0006)	.324 (.0004)	.443 (.0006)
	0.2	.283 (.0004)	.296 (.0004)	.409 (.0005)	.587 (.0006)	.354 (.0005)	.497 (.0006)
	0.4	.333 (.0005)	.306 (.0004)	.415 (.0005)	.589 (.0006)	.391 (.0005)	.551 (.0006)
t_5	0.0	.275 (.0005)	.296 (.0004)	.412 (.0006)	.591 (.0007)	.297 (.0005)	.368 (.0009)
	0.2	.264 (.0006)	.288 (.0004)	.395 (.0005)	.571 (.0006)	.326 (.0007)	.426 (.0011)
	0.4	.331 (.0010)	.301 (.0006)	.398 (.0006)	.564 (.0007)	.370 (.0009)	.494 (.0012)
χ^2 (s)	0.0	.277 (.0005)	.296 (.0004)	.412 (.0006)	.591 (.0007)	.368 (.0008)	.508 (.0012)
	0.2	.337 (.0008)	.314 (.0005)	.431 (.0006)	.608 (.0007)	.408 (.0009)	.562 (.0012)
	0.4	.416 (.0011)	.341 (.0006)	.457 (.0007)	.631 (.0007)	.453 (.0010)	.611 (.0013)
χ^2 (d)	0.0	.275 (.0005)	.297 (.0004)	.412 (.0005)	.591 (.0007)	.381 (.0006)	.540 (.0008)
	0.2	.325 (.0006)	.314 (.0005)	.422 (.0006)	.593 (.0007)	.420 (.0007)	.589 (.0008)
	0.4	.394 (.0008)	.338 (.0005)	.437 (.0006)	.602 (.0007)	.460 (.0007)	.634 (.0008)

Note. R^2 of the latent M and Y equations, averaged over replications, sample sizes, and reliability levels, with the Monte Carlo standard error in parentheses. R^2_M is the same under every (mis)specification. R^2_Y is shown under the correctly specified model (none) and under the small (s) and medium (m) omitted paths c_2 and c_3, which add explained variance to Y. The measurement (mis)specifications leave both values unchanged.

Sensitivity to Outlier Screening

The main text reports performance measures after removing, condition-wide, every replication flagged as an outlier on any estimation approach, that is, an estimate beyond three interquartile ranges from the quartiles or a standard error above 10 on any structural parameter. As a sensitivity check, all measures were recomputed with the screening switched off, retaining every converged replication. Table 3 reports, for each sample size, reliability level, and estimation approach, the proportion of replications removed and the largest absolute difference in each performance measure between the reported value and the value with the removed replications retained, taken over all estimands, predictor distributions, interaction magnitudes, and (mis)specification conditions.

Table 3

Largest Change in Each Performance Measure When Outliers Are Retained

Reliability	*N*	Removed	Estimator	Rel. bias	Rel. RMSE	SE/SD	Coverage	Rejection
Low	100	8.4 (16.2)	LSAM	0.148	0.295	0.163	0.017	0.022
Low	100		LMS	0.281	6.162	0.561	0.019	0.022
Low	100		UPI	1.040	12.653	2.856	0.024	0.011
Low	100		FSR	0.125	0.265	0.195	0.031	0.026
Low	200	3.2 (8.1)	LSAM	0.111	0.350	0.203	0.014	0.017
Low	200		LMS	0.190	0.479	0.259	0.019	0.012
Low	200		UPI	0.989	28.847	1.132	0.016	0.011
Low	200		FSR	0.089	0.306	0.209	0.021	0.015
Low	300	2.1 (6.9)	LSAM	0.075	0.154	0.129	0.016	0.012
Low	300		LMS	0.148	0.424	0.302	0.014	0.012
Low	300		UPI	0.206	3.363	1.293	0.013	0.008
Low	300		FSR	0.059	0.131	0.130	0.020	0.012
Low	500	1.5 (7.7)	LSAM	0.102	0.235	0.281	0.013	0.008
Low	500		LMS	0.153	0.380	0.331	0.010	0.009
Low	500		UPI	0.219	2.266	0.753	0.014	0.008
Low	500		FSR	0.087	0.229	0.304	0.021	0.009
High	100	2.3 (5.8)	LSAM	0.100	0.370	0.161	0.015	0.015
High	100		LMS	0.105	0.393	0.298	0.017	0.017
High	100		UPI	0.179	2.046	2.457	0.018	0.011
High	100		FSR	0.091	0.354	0.154	0.014	0.014
High	200	2.4 (9.4)	LSAM	0.071	0.169	0.128	0.016	0.009
High	200		LMS	0.075	0.175	0.120	0.014	0.007
High	200		UPI	0.113	0.641	0.575	0.017	0.005
High	200		FSR	0.069	0.172	0.130	0.016	0.008
High	300	3.2 (11.1)	LSAM	0.074	0.325	0.245	0.009	0.005
High	300		LMS	0.082	0.333	0.244	0.010	0.004
High	300		UPI	0.083	0.369	0.255	0.008	0.004
High	300		FSR	0.069	0.311	0.213	0.013	0.005
High	500	4.2 (12.2)	LSAM	0.094	0.334	0.306	0.009	0.005
High	500		LMS	0.094	0.335	0.303	0.011	0.004
High	500		UPI	0.105	0.380	0.317	0.011	0.005
High	500		FSR	0.086	0.318	0.292	0.014	0.005

Note. Removed is the mean and, in parentheses, the maximum percentage of replications removed by the outlier screening over the conditions at that sample size and reliability level. Removal is condition-wide, so it is shown once per block. The remaining columns give, for each estimation approach, the largest absolute difference between the reported measure and the measure computed without screening, over all estimands, distributions, interaction magnitudes, and (mis)specification conditions. Rel. = relative. Rejection is the Type I error at $a_3 = 0$ and the power at $a_3 = 0.2$ for the imm and a_3. LSAM = local structural-after-measurement; LMS = latent moderated structural equations; UPI = unconstrained product indicator; FSR = factor score regression.

Extreme Standard Errors and the SE/SD Ratio

The main text reports SE/SD ratios above 1.10 for UPI at low reliability and small sample sizes. To examine whether that overestimation reflects a few replications with very large standard errors rather than a general overstatement of uncertainty, a standard error was classified as extreme when it lay more than three interquartile ranges above the upper quartile of its cell, and the ratio was recomputed with those replications dropped from the mean standard error only. The standard deviation of the estimates keeps every replication, since it is the quantity the standard errors are meant to track. Table 4 lists every condition at $a_3 = 0.2$ under correct specification, the condition shown in the main text, in which the SE/SD ratio of the interaction coefficient or the imm exceeds 1.10.

Table 4

Conditions With an SE/SD Ratio Above 1.10 at $a_3 = 0.2$ Under Correct Specification, With and Without Extreme Standard Errors

Estimand	Estimator	Reliability	N	Distribution	Extreme SEs (%)	SE/SD	SE/SD, removed
a_3	UPI	Low	100	Normal	3.2	1.33	1.14
				Uniform	2.7	1.21	1.09
				t_5	3.0	1.21	1.02
				χ^2 (s)	6.0	1.31	0.94
				χ^2 (d)	3.9	1.54	1.20
			200	χ^2 (s)	5.8	1.25	0.97
				χ^2 (d)	3.4	1.31	1.09
			300	χ^2 (s)	2.5	1.12	1.00
				χ^2 (d)	3.1	1.16	1.04
		High	100	χ^2 (d)	3.2	1.12	0.97
imm	UPI	Low	100	Normal	3.5	1.41	1.20
				Uniform	1.7	1.24	1.15
				t_5	3.0	1.24	1.07
				χ^2 (s)	5.5	1.38	1.00
				χ^2 (d)	4.1	1.64	1.25
			200	Normal	0.9	1.14	1.10
				Uniform	0.7	1.12	1.11
				t_5	1.9	1.10	0.99
				χ^2 (s)	5.3	1.30	1.02
				χ^2 (d)	4.2	1.35	1.09
			300	χ^2 (s)	2.2	1.11	1.00
				χ^2 (d)	2.7	1.20	1.10
		High	100	χ^2 (d)	2.7	1.16	1.02

Note. Extreme SEs is the percentage of replications in the cell whose standard error lies more than three interquartile ranges above the upper quartile. SE/SD is the ratio as reported in the main text, the mean standard error over the standard deviation of the estimates. The last column gives the same ratio after removing the extreme standard errors from the mean, with the standard deviation unchanged. imm = index of moderated mediation; UPI = unconstrained product indicator; (s) = same skewness; (d) = opposite skewness.

Robustness to the Copula

In the main design, the normal condition draws the latent predictors through a Gaussian copula, whereas the four nonnormal conditions combine nonnormal margins with a Clayton copula. The nonnormal conditions therefore differ from the normal condition in the dependence structure as well as in the margins. To separate the two, the four nonnormal conditions were regenerated with a Gaussian copula, so that only the margins differ from the normal condition, with every other aspect of the design, including the calibrated misspecification magnitudes, held fixed. Table 5 reports, for each sample size, reliability level, and nonnormal distribution, the largest absolute difference in each performance measure between the Clayton and the Gaussian copula results.

Table 5

Largest Change in Each Performance Measure Under a Gaussian Instead of a Clayton Copula

`Pending. Run the robustness simulation (copula_robustness = TRUE) and analyze_mc.R.`

Note. Each entry is the largest absolute difference between the measure under the Clayton copula (main design) and under the Gaussian copula, over all estimation approaches, estimands, interaction magnitudes, and (mis)specification conditions at that sample size, reliability level, and distribution. Rel. = relative. Rejection is the Type I error at $a_3 = 0$ and the power at $a_3 = 0.2$ for the imm and a_3.

Complete Figures

The figures below show the complete results from which the figures in the main text were drawn. In each, LSAM = local structural-after-measurement, LMS = latent moderated structural equations, UPI = unconstrained product indicator, FSR = factor score regression, imm = index of moderated mediation, and n = no misspecification. Where shown, the gray band marks the acceptable region for the measure and the error bars represent the Monte Carlo standard error (MCSE).

Figure 1

Relative Bias of the imm (a_3b) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions

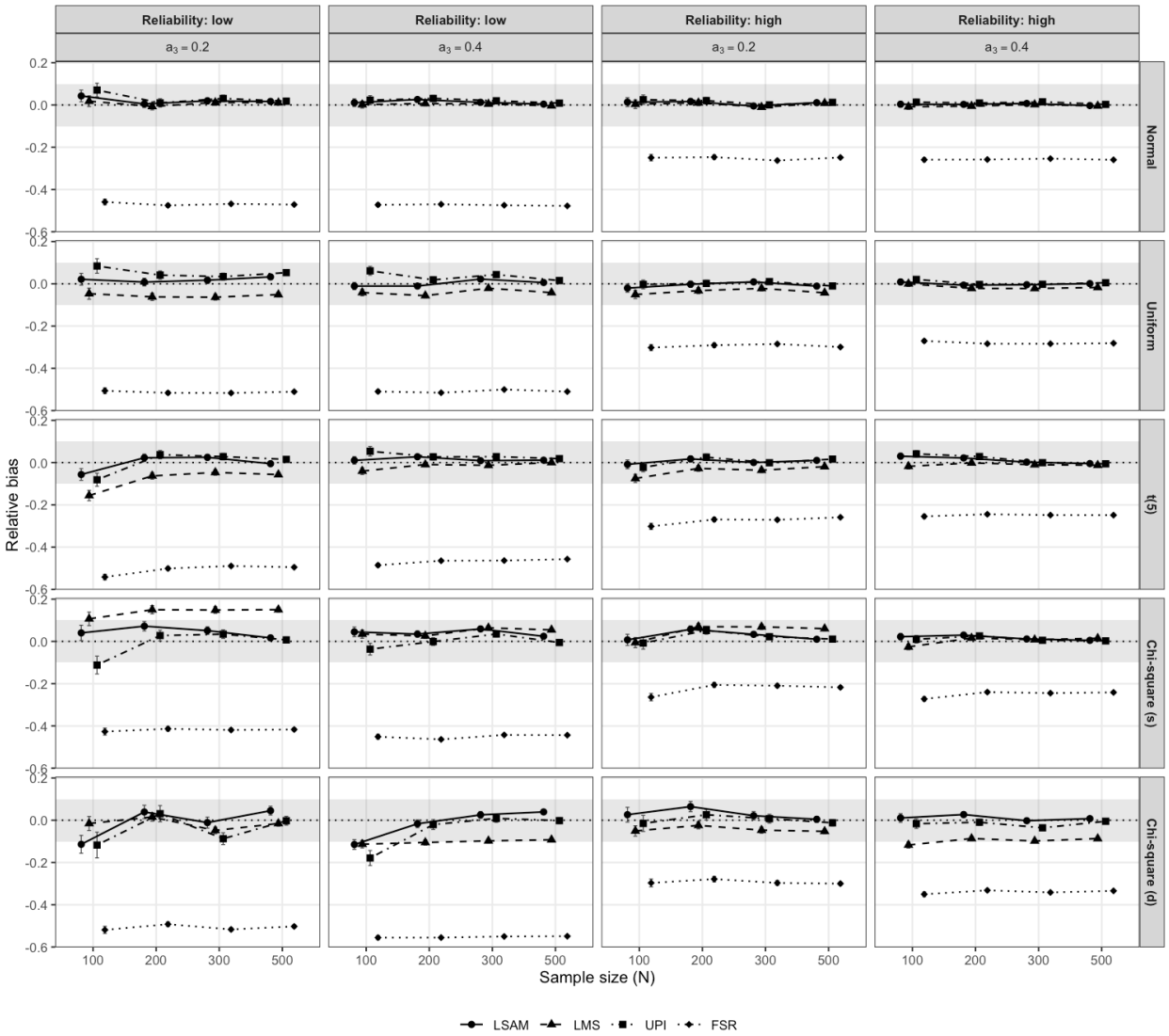


Figure 2

Relative Bias of the Interaction Coefficient (a_3) Under the Correctly Specified Condition Across Sample Sizes, Reliability, Interaction Values, and Distributions

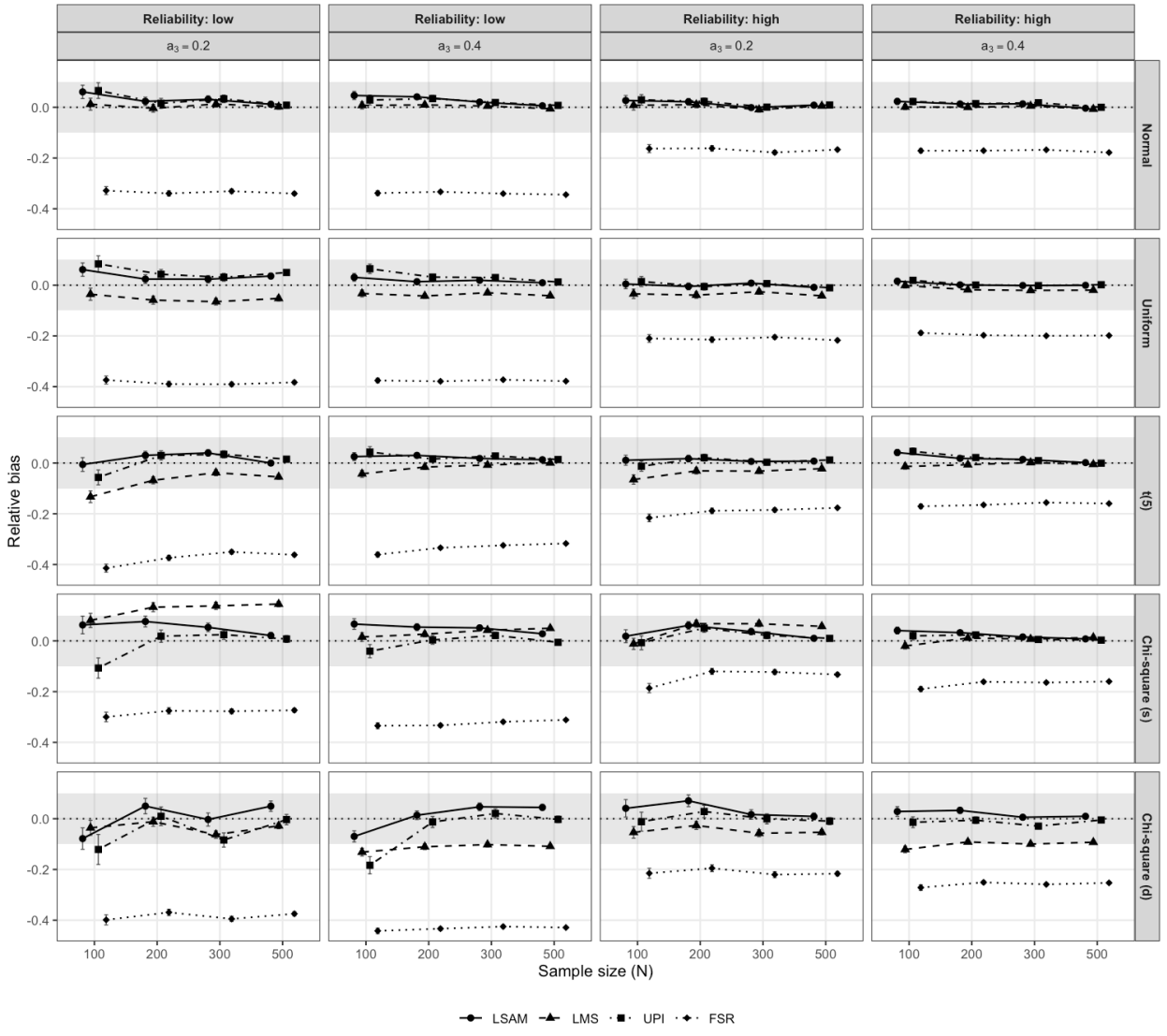


Figure 3

Relative Bias of the imm ($a_3 b$) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

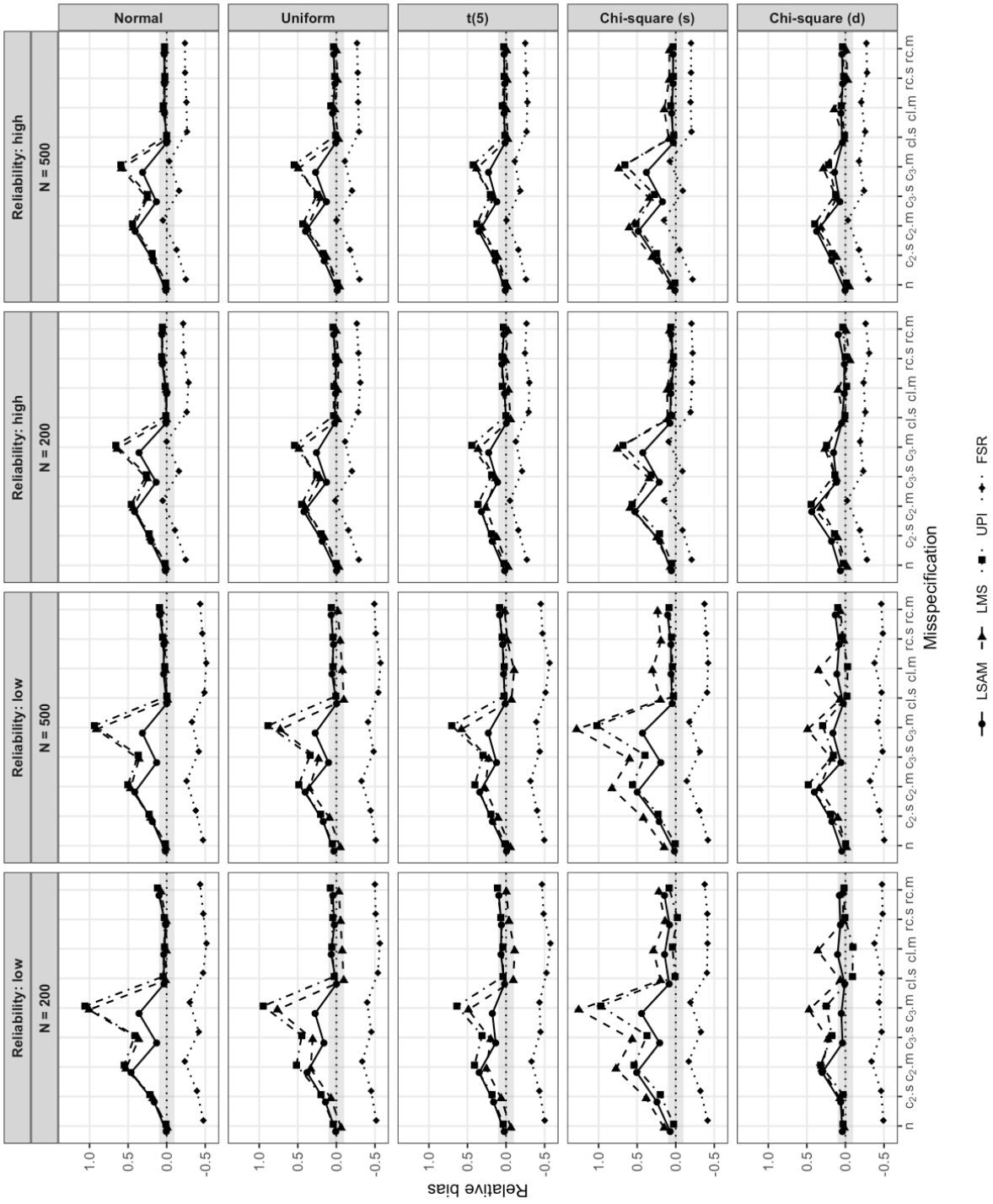


Figure 4

Relative Bias of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

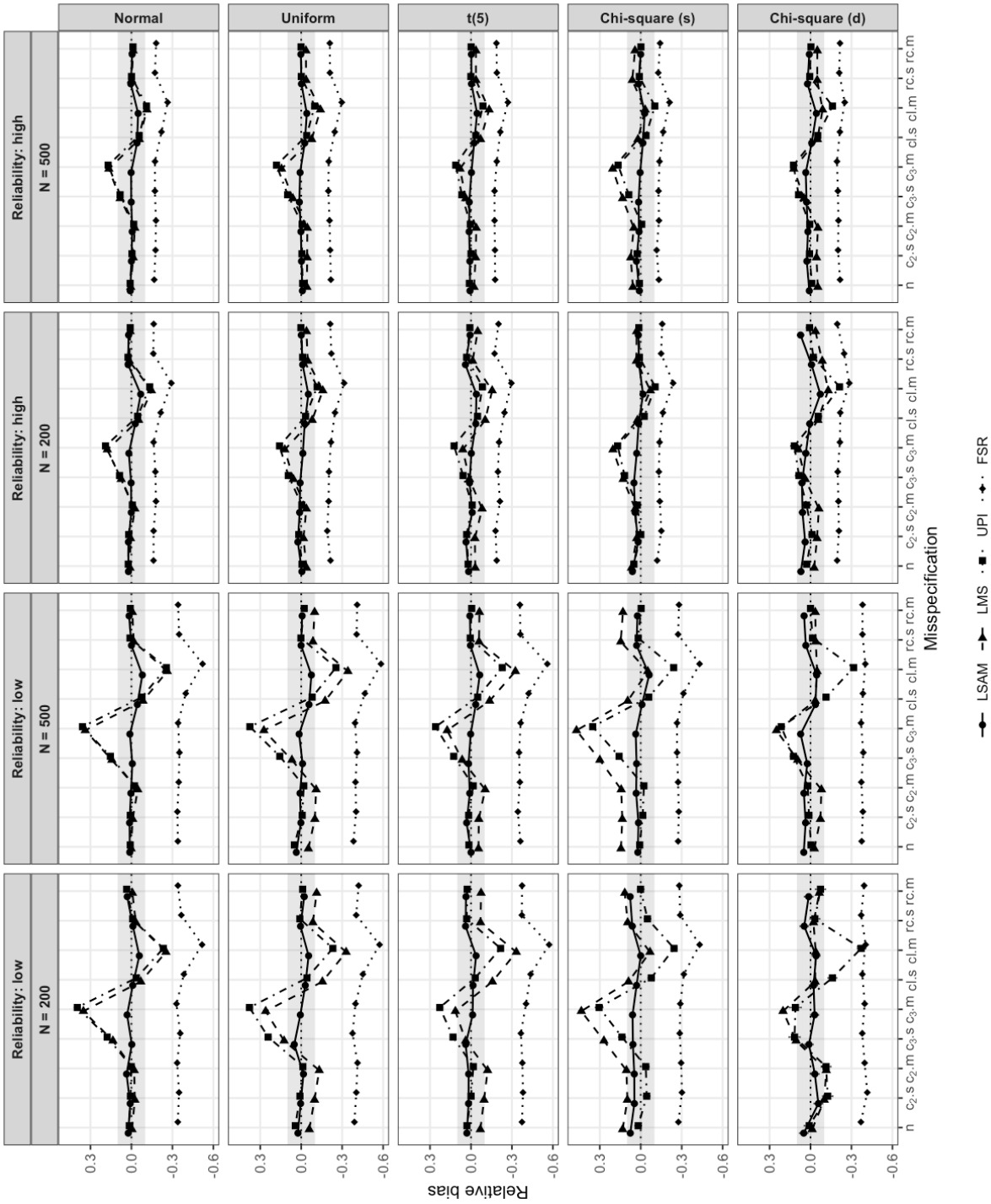


Figure 5

Relative RMSE of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

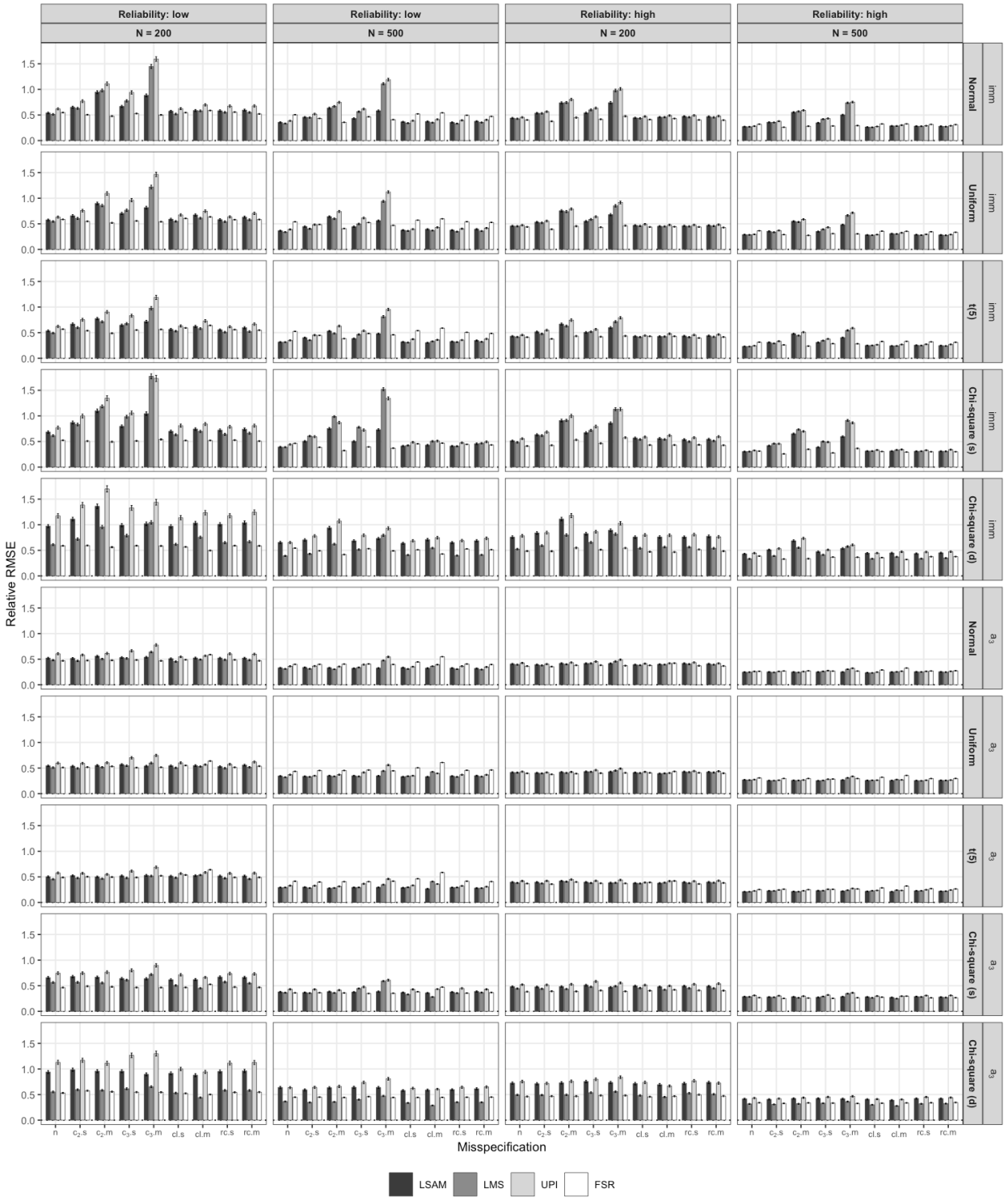


Figure 6

SE/SD of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

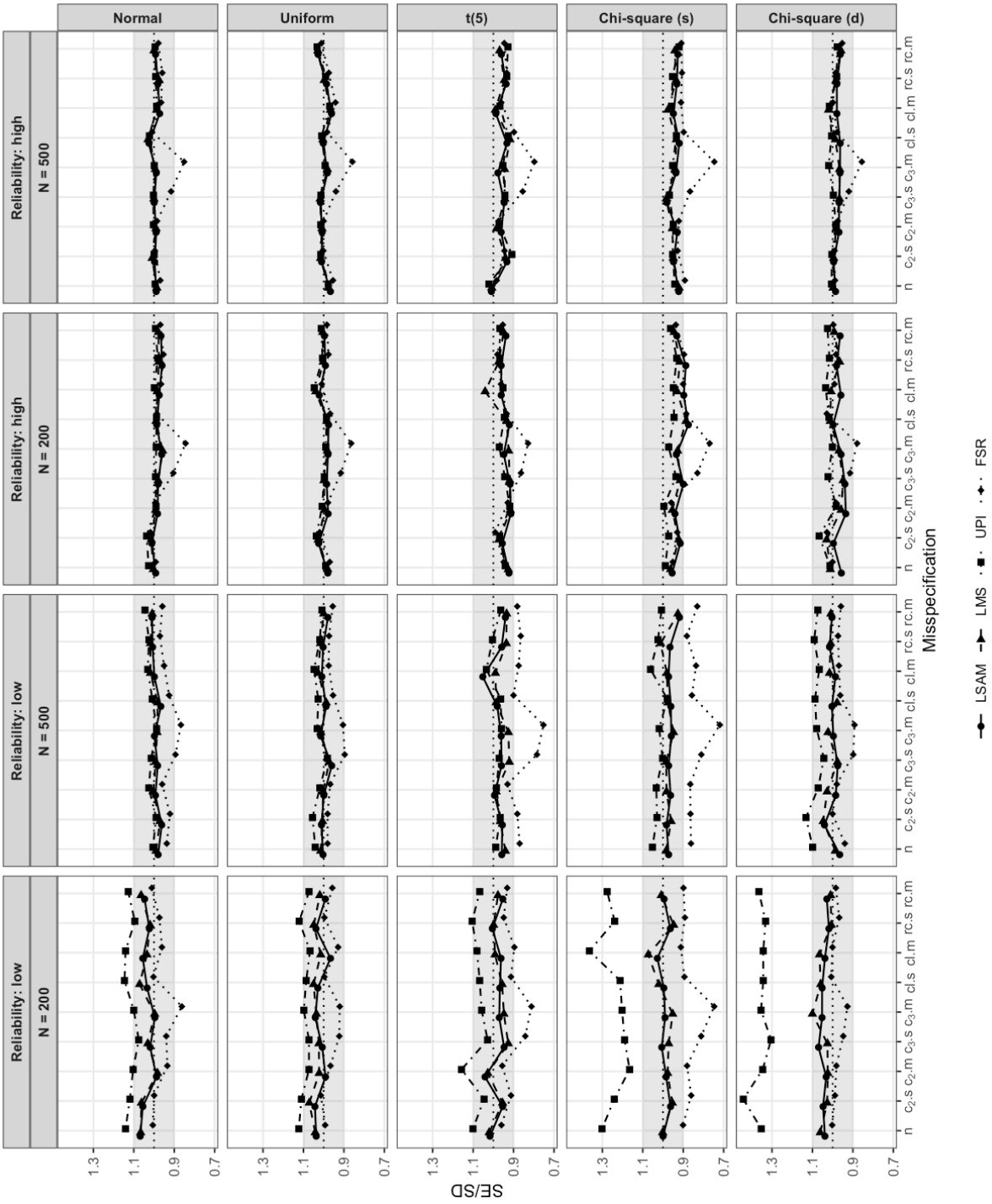


Figure 7

SE/SD of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for N = 200 and 500 at a_3 = 0.2

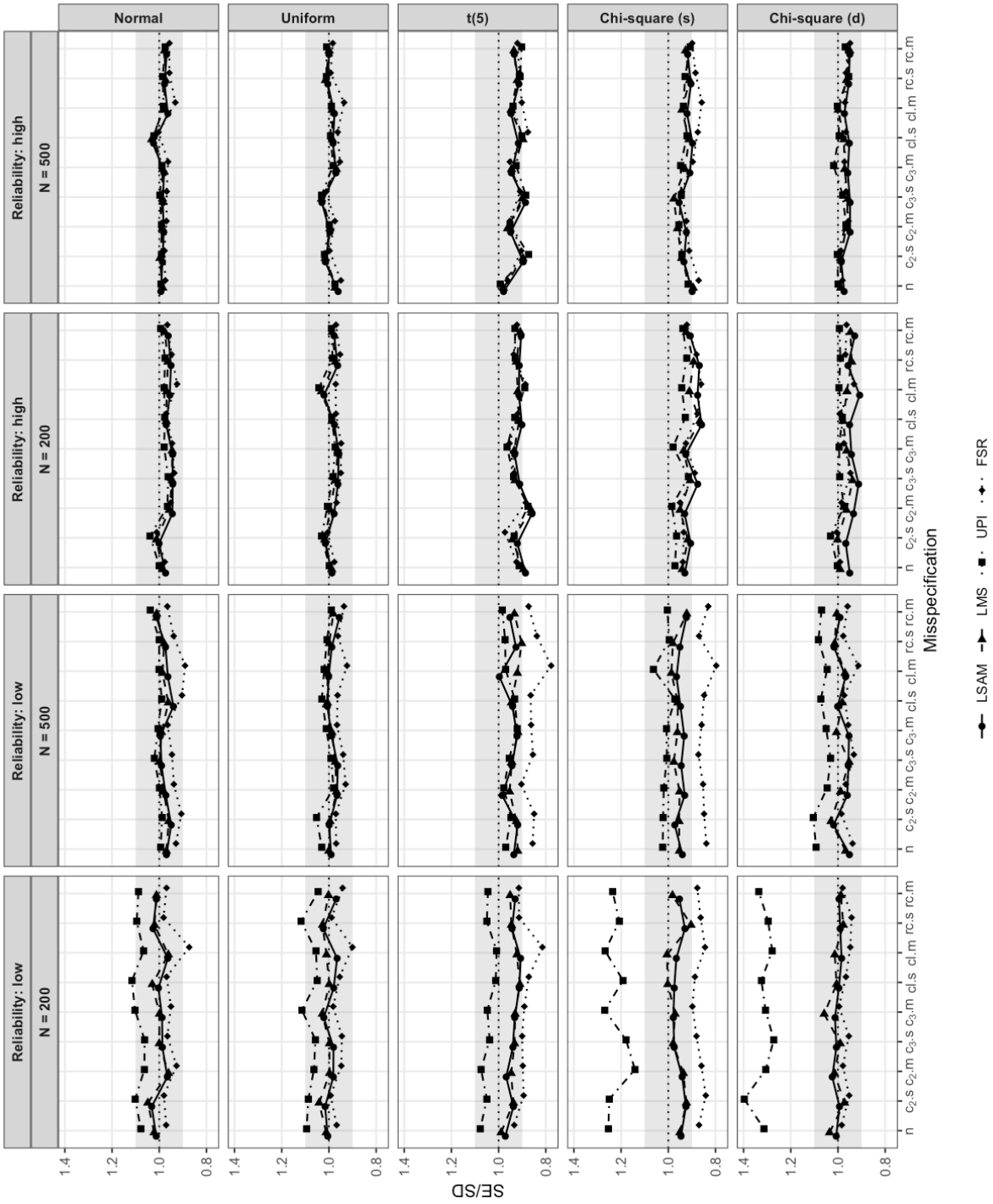


Figure 8

Coverage of the imm (a_3b) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

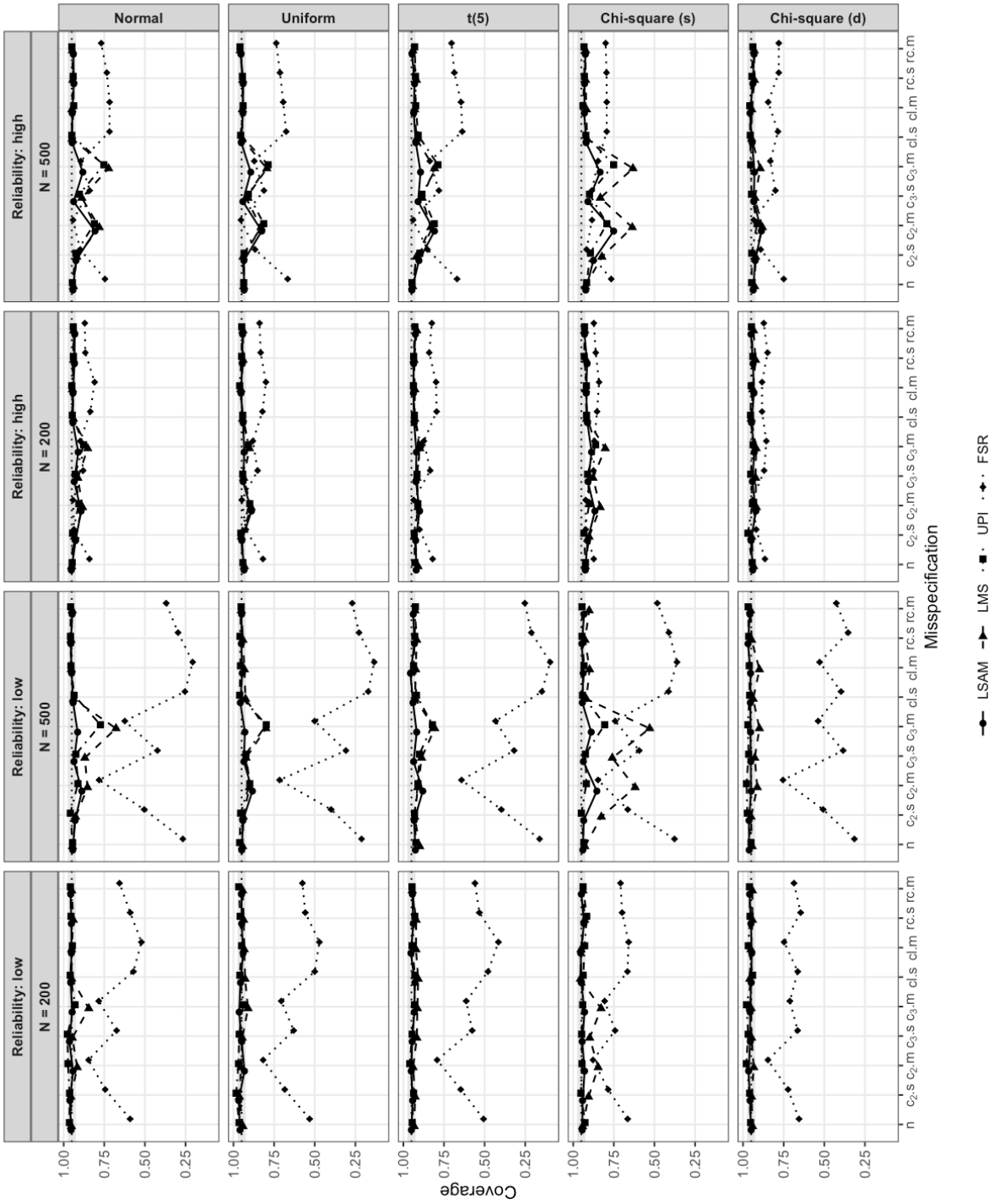


Figure 9

Coverage of the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

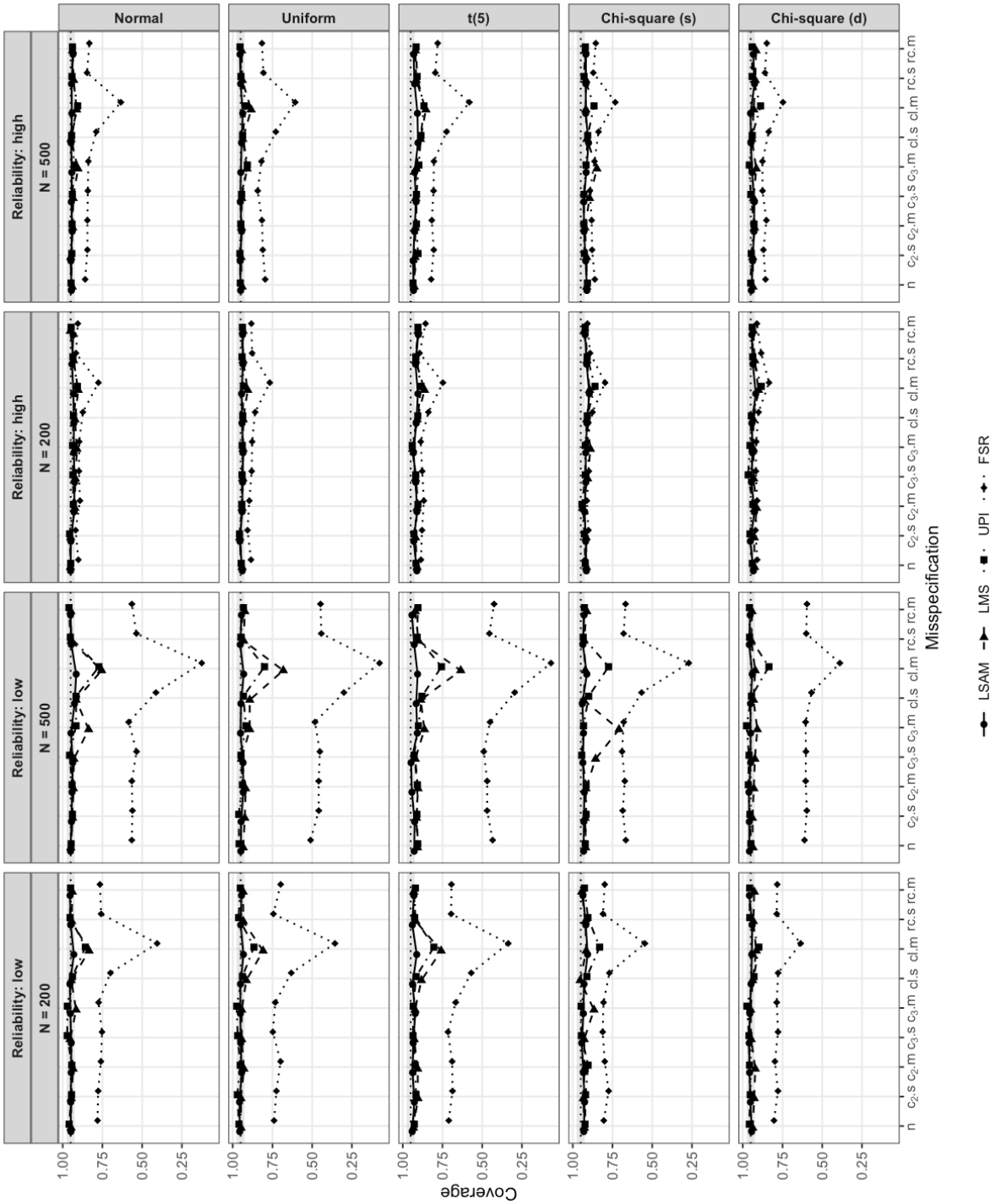


Figure 10

Type I Error of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0$

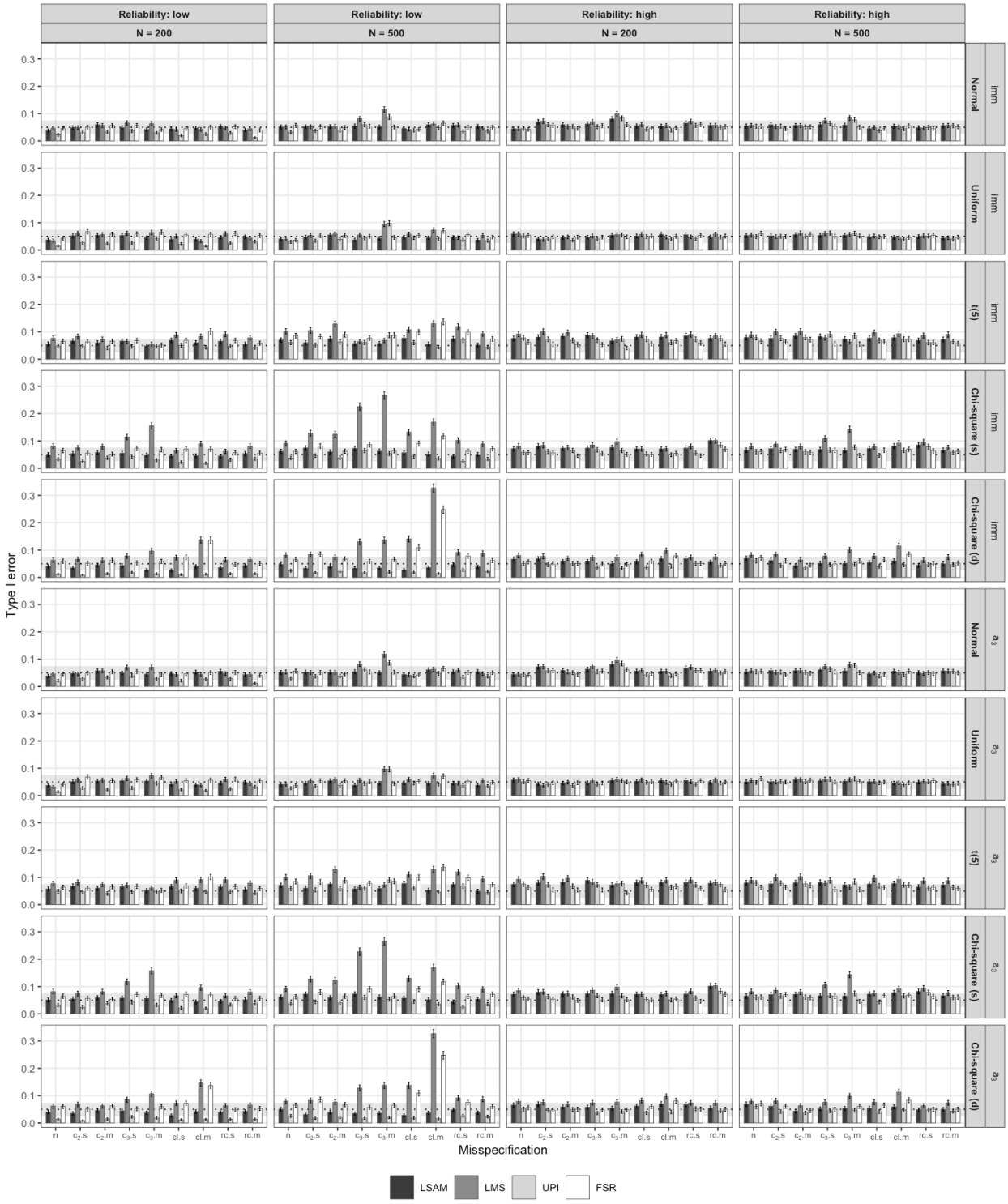


Figure 11

Power for the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions, Reliability, and (Mis)specifications for $N = 200$ and 500 at $a_3 = 0.2$

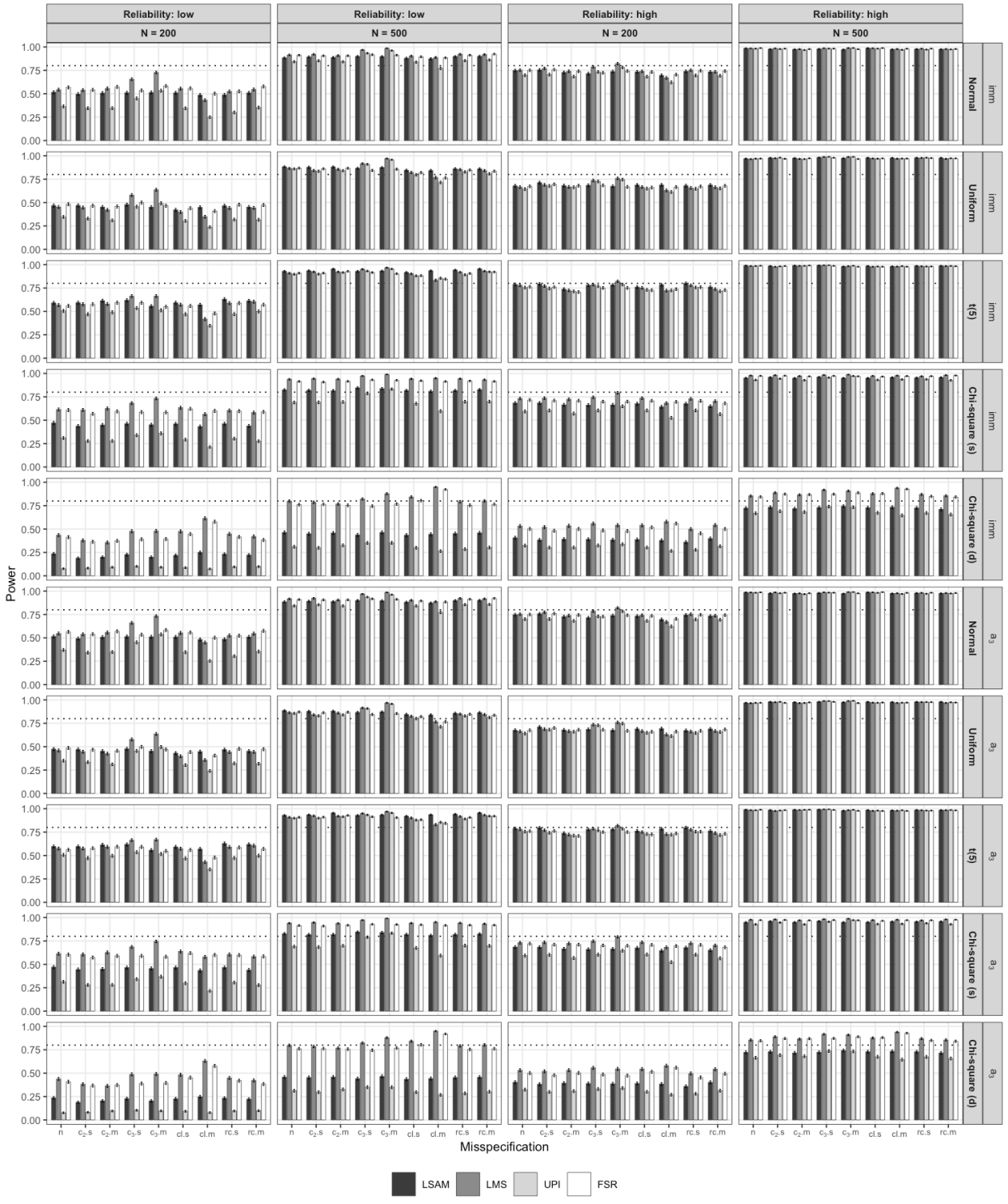


Figure 12

Relative Bias of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$

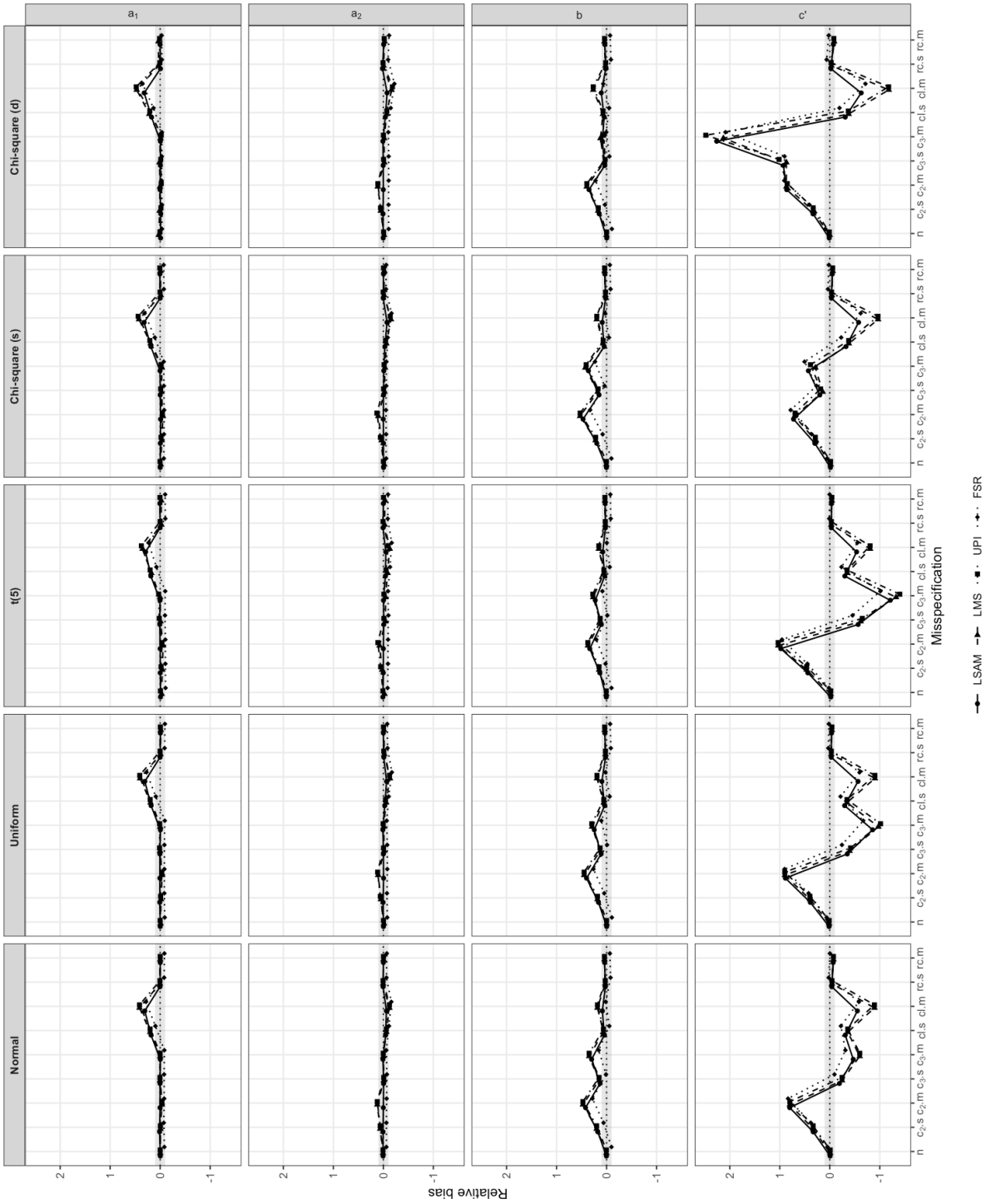


Figure 13

Coverage of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$

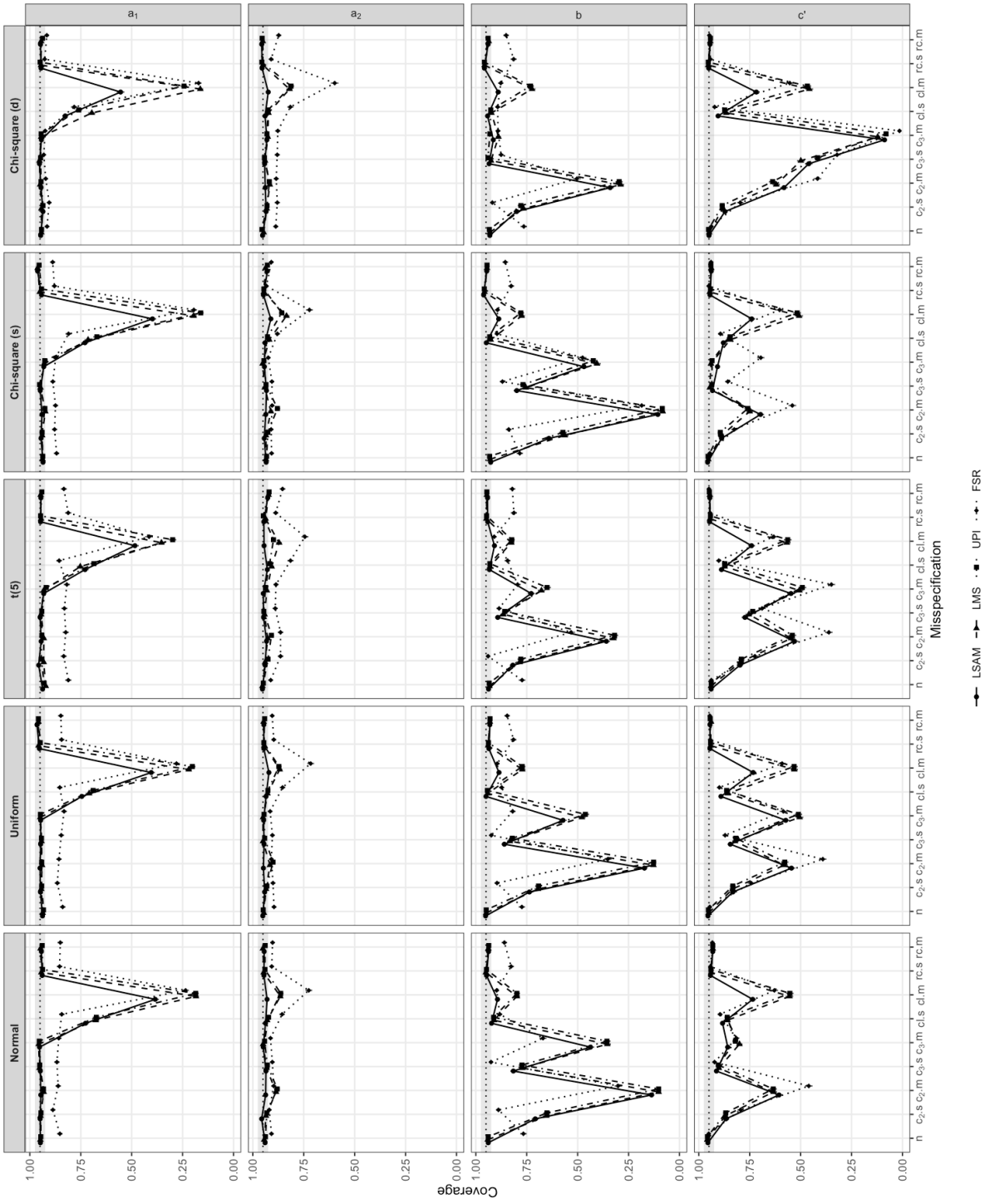


Figure 14

SE/SD of the Structural Coefficients (a_1, a_2, b, c') Across Distributions and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$

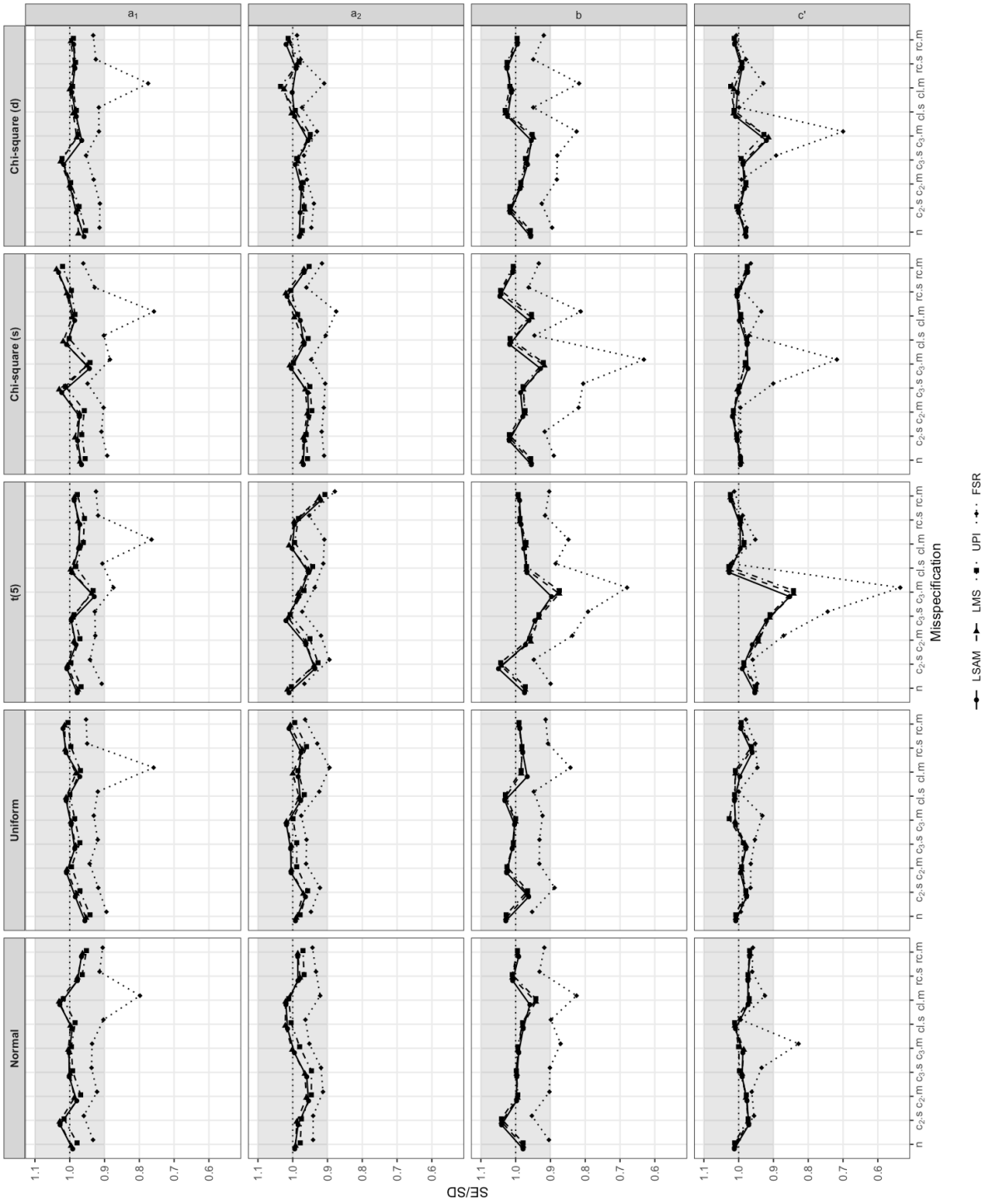


Figure 15

SE^2/VAR Ratio of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions and (Mis)specifications at High Reliability for $N = 200$ and 500 at $a_3 = 0.2$

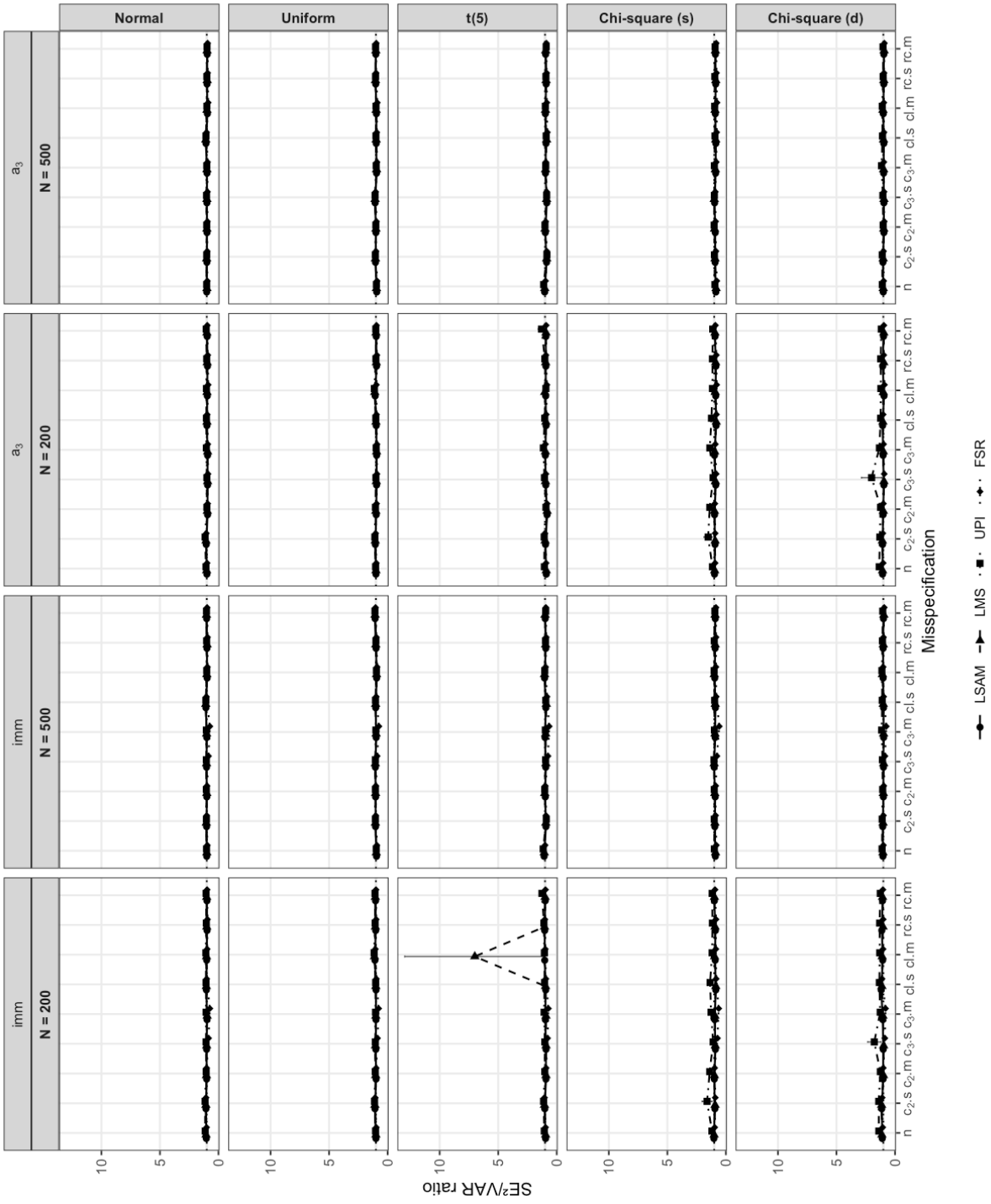


Figure 16

SE^2/VAR Ratio of the imm (a_3b) and the Interaction Coefficient (a_3) Across Distributions and (Mis)specifications at Low Reliability for $N = 200$ and 500 at $a_3 = 0.2$

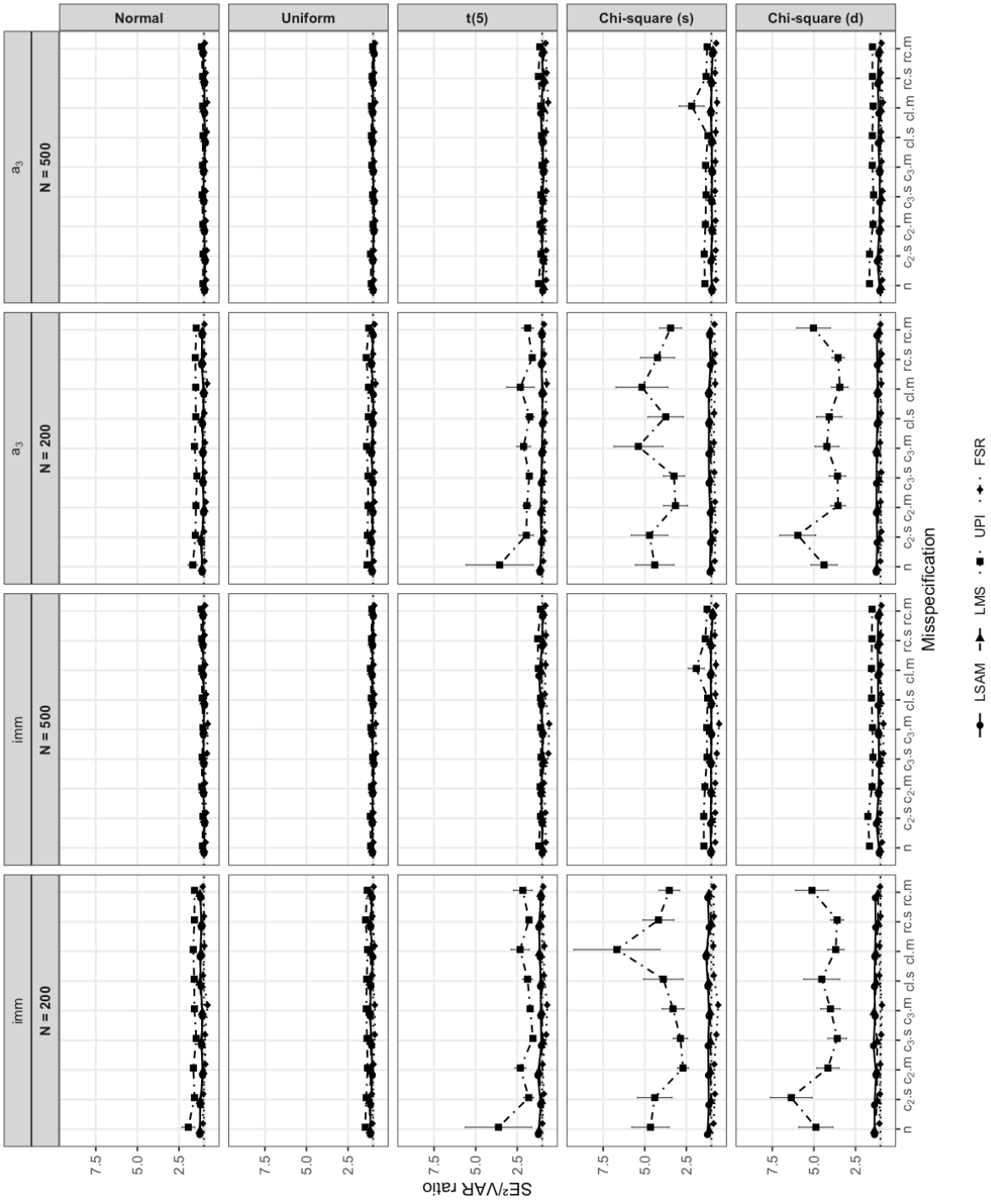


Figure 17

Performance of the Cross-Loaded Mediator Indicator Loading (m_3) Across All Measures, Distributions, and (Mis)specifications at High Reliability, $N = 500$, and $a_3 = 0.2$

